

# SmartApeSG Fake Browser Update Investigation

NetSupport RAT & StealC Infection Analysis

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## Contents

|                                                                           |           |
|---------------------------------------------------------------------------|-----------|
| <b>1. Executive Summary</b>                                               | <b>2</b>  |
| <b>2. Investigation Overview</b>                                          | <b>2</b>  |
| <b>3. Investigation Scope</b>                                             | <b>2</b>  |
| <b>4. Investigation Details</b>                                           | <b>3</b>  |
| <b>5. Key Findings</b>                                                    | <b>3</b>  |
| 5.1 Suspicious DNS Activity . . . . .                                     | 3         |
| 5.2 Command-and-Control Communication . . . . .                           | 4         |
| 5.3 Encrypted Traffic Analysis . . . . .                                  | 4         |
| 5.4 Data Transfer Patterns . . . . .                                      | 5         |
| 5.5 Threat Intelligence Validation . . . . .                              | 7         |
| <b>6. Indicators of Compromise (IOCs)</b>                                 | <b>8</b>  |
| Observed Indicators (High Confidence) . . . . .                           | 8         |
| Internal Host: . . . . .                                                  | 8         |
| IP Addresses: . . . . .                                                   | 8         |
| Domains: . . . . .                                                        | 9         |
| Correlated Indicators (Medium Confidence – Threat Intelligence) . . . . . | 9         |
| IP Addresses: . . . . .                                                   | 9         |
| Network Indicators (URLs / Endpoints) . . . . .                           | 9         |
| Initial Infection: . . . . .                                              | 9         |
| NetSupport RAT Activity: . . . . .                                        | 9         |
| StealC Command-and-Control: . . . . .                                     | 9         |
| Payload Delivery: . . . . .                                               | 9         |
| Behavioral Indicators (Detection-Oriented) . . . . .                      | 10        |
| <b>7. MITRE ATT&amp;CK Mapping (Observed Activity)</b>                    | <b>10</b> |
| ATT&CK Summary . . . . .                                                  | 11        |
| <b>8. Recommendations</b>                                                 | <b>11</b> |
| <b>9. Conclusion &amp; Risk Assessment</b>                                | <b>12</b> |

## 1. Executive Summary

Network traffic analysis identified a compromised internal host communicating with external infrastructure associated with malicious activity. The behavior is consistent with a Remote Access Trojan (NetSupport RAT) following a user-driven infection chain involving a fake browser update. The infected system established repeated outbound connections over encrypted channels, enabling persistent remote access and control by an external actor.

Analysis also revealed significant inbound data transfers, indicating the delivery of additional payloads and potential staging of information-stealing malware (StealC). The use of encrypted communication and legitimate-looking domains allowed the activity to blend with normal traffic, increasing the likelihood of evasion. This activity represents a **high-risk security incident**, as it provides attackers with sustained access and the capability to exfiltrate sensitive data. Immediate containment, credential rotation, and full host remediation are recommended to prevent further attacker access and potential data loss.

## 2. Investigation Overview

This analysis was conducted on a network traffic capture from the **2025-02-18 SmartApeSG infection scenario**, involving a fake browser update leading to NetSupport RAT and StealC malware. The dataset was sourced from Malware-Traffic-Analysis.net and analyzed using Zeek (log-based analysis), Wireshark (packet inspection), NetworkMiner (artifact extraction), and VirusTotal (IOC validation).

The objective was to identify malicious activity, map command-and-control communication, and extract actionable indicators of compromise.

Source: <https://www.malware-traffic-analysis.net/2025/02/18/index.html>

## 3. Investigation Scope

The investigation focused on analysis of network traffic and associated Zeek logs generated from the provided PCAP dataset.

The scope included:

- DNS activity analysis
- Command-and-control identification
- TLS communication analysis
- HTTP activity review
- Artifact extraction

- IOC validation
- MITRE ATT&CK mapping

Host-based forensic analysis was outside the scope of this investigation.

## 4. Investigation Details

| Case Overview                                                    |
|------------------------------------------------------------------|
| <b>Investigation Type:</b> Network Traffic Analysis              |
| <b>Malware Family:</b> NetSupport RAT, StealC                    |
| <b>Data Source:</b> PCAP                                         |
| <b>Analysis Tools:</b> Zeek, Wireshark, NetworkMiner, VirusTotal |
| <b>Infected Host:</b> 10.2.18.101                                |
| <b>Primary C2 Infrastructure:</b> 79.141.161.169                 |
| <b>Severity:</b> High                                            |

## 5. Key Findings

### 5.1 Suspicious DNS Activity

Analysis of DNS logs revealed multiple queries from the internal host **10.2.18.101** to suspicious and contextually relevant domains, including:

- **cinaweine.shop**
- **poormet.com**
- **geo.netsupportsoftware.com**

The domain **cinaweine.shop** is particularly notable due to its uncommon naming pattern and repeated resolution behavior, indicating active use during the infection chain.

The domain **poormet.com** is associated with infrastructure used for payload delivery and is fronted by Cloudflare, a common technique to obscure backend infrastructure.

Additionally, **geo.netsupportsoftware.com** is a legitimate domain leveraged by NetSupport RAT, aligning with known malware behavior.

**Implication:** Indicates the use of DNS for resolving attacker-controlled infrastructure and supporting malware communication.

```
analyst@lab-vm:~/zeek-lab$ cat dns.log | zeek-cut ts id.orig_h query | grep -E "cinaweine|poormet|netsupport"
1739896569.274890 10.2.18.101 cinaweine.shop
1739896570.283934 10.2.18.101 cinaweine.shop
1739896629.836845 10.2.18.101 cinaweine.shop
1739896630.975336 10.2.18.101 poormet.com
1739896634.472518 10.2.18.101 geo.netsupportsoftware.com
```

**Figure 1:** Zeek DNS log showing host 10.2.18.101 resolving suspicious domains associated with the infection chain, including cinaweine.shop, poormet.com, and geo.netsupportsoftware.com.

## 5.2 Command-and-Control Communication

Connection analysis identified consistent outbound communication from the infected host **10.2.18.101** to the external IP address **79.141.161.169** over **TCP port 443**. The traffic pattern demonstrates repeated connections occurring at short and regular intervals, characteristic of automated beaconing behavior.

This communication was not associated with standard web browsing activity and persisted over time, suggesting an established command-and-control (C2) channel. The consistency in destination IP and port further reinforces the likelihood of attacker-controlled infrastructure.

**Implication:** Indicates persistent beaconing to a command-and-control server, enabling remote access and control of the compromised host.

```
analyst@lab-vm:~/zeek-lab$ cat conn.log | zeek-cut ts id.orig_h id.resp_h id.resp_p | grep "79.141.161.169"
1739896570.501385 10.2.18.101 79.141.161.169 443
1739896571.884143 10.2.18.101 79.141.161.169 443
1739896573.740359 10.2.18.101 79.141.161.169 443
1739896573.742038 10.2.18.101 79.141.161.169 443
1739896573.743701 10.2.18.101 79.141.161.169 443
1739896573.742427 10.2.18.101 79.141.161.169 443
1739896629.868793 10.2.18.101 79.141.161.169 443
```

**Figure 2:** Zeek connection log showing repeated outbound connections from host 10.2.18.101 to external IP address 79.141.161.169 over TCP port 443. The recurring connection pattern is consistent with command-and-control beaconing activity associated with NetSupport RAT.

## 5.3 Encrypted Traffic Analysis

Analysis of TLS logs confirmed that communication with the identified external infrastructure occurred over **TLSv1.2** and **TLSv1.3**. Notably, there was an absence of corresponding entries in HTTP logs, despite traffic occurring over port 443.

This suggests that the communication is fully encrypted using HTTPS or utilizes TLS-based communication that prevents visibility into application-layer content. This behavior is consistent with modern Remote Access Trojans that rely on encrypted communication to evade network-based detection.

**Implication:** Indicates the use of encrypted channels to conceal command-and-control communication and evade network-based detection mechanisms.

```
analyst@lab-vn:~/zeek-lab$ cat ssl.log | zeek-cut ts id.orig_h id.resp_h version | grep "79.141.161.169"
1739896570.765656 10.2.18.101 79.141.161.169 TLSv12
1739896572.344090 10.2.18.101 79.141.161.169 TLSv12
1739896573.906230 10.2.18.101 79.141.161.169 TLSv12
1739896573.905868 10.2.18.101 79.141.161.169 TLSv12
1739896573.905972 10.2.18.101 79.141.161.169 TLSv12
1739896573.906023 10.2.18.101 79.141.161.169 TLSv12
1739896629.921968 10.2.18.101 79.141.161.169 TLSv13
```

**Figure 3:** Zeek TLS log showing encrypted communications between host 10.2.18.101 and external IP address 79.141.161.169 using TLSv1.2 and TLSv1.3. The use of encrypted channels limited visibility into application-layer content and is consistent with command-and-control traffic designed to evade network inspection.

## 5.4 Data Transfer Patterns

Analysis of connection byte counts revealed a consistent pattern of relatively small outbound requests from the infected host followed by significantly larger inbound responses from the external server.

Examples include:

- ~1 KB outbound → hundreds of KB to multiple MB inbound responses

This asymmetry suggests that the infected system is issuing lightweight requests (beacons or tasking signals) and receiving larger payloads, which may include commands, modules, or additional malware components such as StealC.

**Implication:** Indicates active data exchange consistent with command delivery and potential payload staging from attacker-controlled infrastructure.

```
analyst@lab-vm:~/zeek-lab$ cat conn.log | zeek-cut ts id.orig_h id.resp_h id.resp_p orig_bytes resp_bytes | grep "79.141.161.169"
```

| ts                | id.orig_h   | id.resp_h      | id.resp_p | orig_bytes | resp_bytes |
|-------------------|-------------|----------------|-----------|------------|------------|
| 1739896570.501385 | 10.2.18.101 | 79.141.161.169 | 443       | 885        | 9012       |
| 1739896571.884143 | 10.2.18.101 | 79.141.161.169 | 443       | 1037       | 348494     |
| 1739896573.740359 | 10.2.18.101 | 79.141.161.169 | 443       | 889        | 1221       |
| 1739896573.742038 | 10.2.18.101 | 79.141.161.169 | 443       | 955        | 1048       |
| 1739896573.743701 | 10.2.18.101 | 79.141.161.169 | 443       | 905        | 1442       |
| 1739896573.742427 | 10.2.18.101 | 79.141.161.169 | 443       | 1676       | 2364434    |
| 1739896629.868793 | 10.2.18.101 | 79.141.161.169 | 443       | 768        | 6013       |

**Figure 4:** Zeek connection log showing a substantial imbalance between outbound and inbound byte counts during communication with 79.141.161.169. Small outbound requests (orig\_bytes) followed by significantly larger inbound responses (resp\_bytes) are consistent with command-and-control tasking and payload delivery behavior.

Additionally, HTTP POST requests to external infrastructure were observed, indicating active communication with attacker-controlled endpoints.

```
analyst@lab-wm:~/zeek-lab$ grep -v '^#' http.log | grep POST | awk '{print "SRC:", $3, "DST:", $5, "METHOD:", $8, "HOST:", $9, "URI:", $10}' | head
```

**Figure 5:** Zeek HTTP log showing POST requests from infected host 10.2.18.101 to attacker-controlled infrastructure 62.164.130.69. The requests target the endpoint /651b5330b08aff3e.php, consistent with command-and-control communications and potential StealC data exfiltration activity.

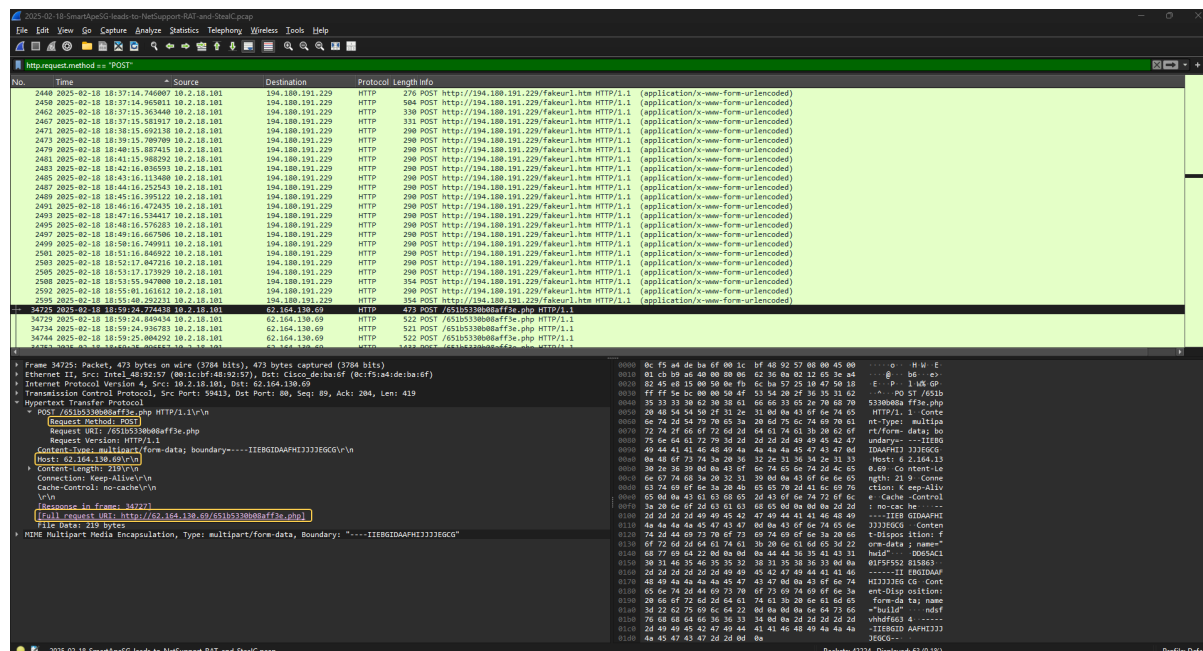
Further validation using NetworkMiner confirmed the transfer of multiple artifacts, including DLL files associated with payload delivery as well as text-based data consistent with browser information, system profiling, and credential harvesting, supporting evidence of post-compromise activity and likely data exfiltration.

[illegible]

**Figure 6:** NetworkMiner artifact extraction showing inbound DLL payload delivery from attacker infrastructure (62.164.130.69 → 10.2.18.101) and outbound transmission of system information, browser data, and credential-related files (10.2.18.101 → 62.164.130.69). The recovered artifacts are consistent with

## StealC information theft and data exfiltration activity.

To further validate this behavior at the packet level, inspection was performed using Wireshark.



**Figure 7:** Wireshark packet analysis showing HTTP POST requests from infected host 10.2.18.101 to attacker-controlled infrastructure 62.164.130.69. Inspection of the HTTP request confirms communication with the endpoint `/651b5330b08aff3e.php`, validating command-and-control activity and potential StealC-related data exfiltration observed in Zeek and NetworkMiner analysis.

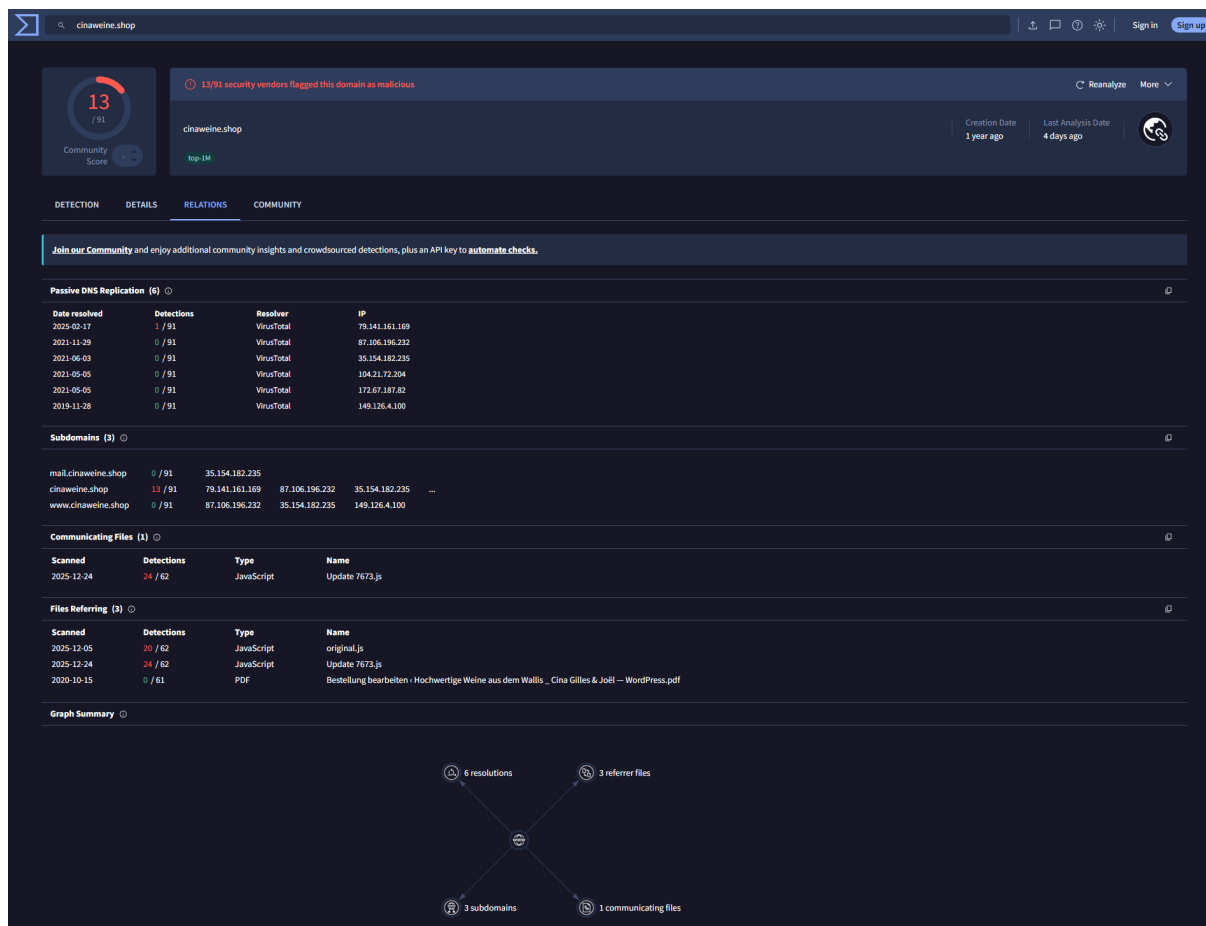
## 5.5 Threat Intelligence Validation

Threat intelligence enrichment was performed using VirusTotal to validate indicators identified during network analysis.

The domain `cinaweine.shop`, observed throughout the infection chain, was associated with multiple malicious detections and infrastructure linked to the investigation. VirusTotal relationships further connected the domain to external IP address `79.141.161.169` and malicious files including `Update7673.js`, increasing confidence that the domain was used as part of the NetSupport RAT and StealC infection infrastructure.

**Implication:** Independent threat intelligence corroborates the malicious nature of infrastructure identified during network analysis.





**Figure 8:** VirusTotal threat intelligence enrichment for cinaweine.shop showing malicious detections, related infrastructure (79.141.161.169), and associated malicious files observed during the investigation.

## 6. Indicators of Compromise (IOCs)

### Observed Indicators (High Confidence)

Internal Host:

- 10.2.18.101

IP Addresses:

- 79.141.161.169
- 62.164.130.69

**Domains:**

- cinaweine.shop
- geo.netsupportsoftware.com
- poormet.com

**Correlated Indicators (Medium Confidence – Threat Intelligence)****IP Addresses:**

- 194.180.191.229

**Network Indicators (URLs / Endpoints)****Initial Infection:**

- <https://cinaweine.shop/work/>

**NetSupport RAT Activity:**

- <http://geo.netsupportsoftware.com/location/loca.asp>
- <http://194.180.191.229/fakeurl.htm>

**StealC Command-and-Control:**

- <http://62.164.130.69/>
- <http://62.164.130.69/651b5330b08aff3e.php>

**Payload Delivery:**

- <http://62.164.130.69/16fcfdf0c5b3315a/sqlite3.dll>
- <http://62.164.130.69/16fcfdf0c5b3315a/freebl3.dll>
- <http://62.164.130.69/16fcfdf0c5b3315a/mozglue.dll>
- <http://62.164.130.69/16fcfdf0c5b3315a/msvc140.dll>
- <http://62.164.130.69/16fcfdf0c5b3315a/nss3.dll>
- <http://62.164.130.69/16fcfdf0c5b3315a/softoken3.dll>
- <http://62.164.130.69/16fcfdf0c5b3315a/vcruntime140.dll>

## Behavioral Indicators (Detection-Oriented)

- POST requests to PHP endpoints using multipart/form-data
- Repeated outbound connections over TCP 443 to a consistent external IP
- Encrypted command-and-control communication without corresponding HTTP visibility
- Large inbound data transfers following small outbound requests
- Retrieval of multiple DLL files from external infrastructure
- Evidence of system and browser data collection with potential exfiltration

## 7. MITRE ATT&CK Mapping (Observed Activity)

Observed malicious activity was mapped to the MITRE ATT&CK framework to contextualize attacker behavior and identify relevant techniques used during the compromise.

| Observed Behavior                               | Supporting Evidence                                               | MITRE Technique                      |
|-------------------------------------------------|-------------------------------------------------------------------|--------------------------------------|
| Suspicious DNS queries to external domains      | dns.log (cinaweine.shop, poormet.com, geo.netsupportsoftware.com) | T1071.004 – DNS                      |
| Command-and-control beaconing over TCP 443      | conn.log (repeated connections to 79.141.161.169)                 | T1071.001 – Web Protocols            |
| Encrypted communication over TLS                | ssl.log (TLSv1.2 / TLSv1.3, no HTTP visibility)                   | T1573 – Encrypted Channel            |
| Large inbound data transfers from external host | conn.log (byte imbalance patterns)                                | T1105 – Ingress Tool Transfer        |
| Data exfiltration via HTTP POST requests        | http.log (POST to /651b5330b08aff3e.php)                          | T1041 – Exfiltration Over C2 Channel |
| Multi-stage payload delivery (DLL retrieval)    | http.log (multiple DLL downloads from external server)            | T1105 – Ingress Tool Transfer        |

## ATT&CK Summary

| ATT&CK ID | Technique                    | Evidence                                      |
|-----------|------------------------------|-----------------------------------------------|
| T1105     | Ingress Tool Transfer        | DLL retrieval from external infrastructure    |
| T1071     | Application Layer Protocol   | HTTP/TLS command-and-control traffic          |
| T1041     | Exfiltration Over C2 Channel | HTTP POST requests to attacker infrastructure |
| T1573     | Encrypted Channel            | TLSv1.2/TLSv1.3 communications                |

The observed activity aligns with common malware post-compromise behaviors, including command-and-control communication, payload delivery, encrypted communications, and data exfiltration. The combination of these techniques is consistent with NetSupport RAT and StealC malware operations.

## 8. Recommendations

The following actions are recommended:

- Immediately isolate affected host 10.2.18.101
- Reset credentials associated with the compromised system
- Block identified malicious IP addresses and domains
- Review proxy and DNS logs for additional affected hosts
- Deploy detections based on identified IOCs
- Hunt for NetSupport RAT and StealC indicators across the environment
- Monitor for additional outbound communication to identified infrastructure
- Develop detections based on identified ATT&CK techniques and observed IOC patterns

## 9. Conclusion & Risk Assessment

The analysis confirms that the internal host **10.2.18.101** was compromised and engaged in active command-and-control (C2) communication with external attacker-controlled infrastructure. The observed behavior is consistent with a **Remote Access Trojan (NetSupport RAT)** followed by secondary activity associated with **information-stealing malware (StealC)**. Evidence includes persistent beaconing to an external IP over TCP 443, use of encrypted channels to evade inspection, and significant inbound data transfers indicative of command delivery and payload staging. This confirms that the system was under active remote control during the observed timeframe.

Additionally, HTTP POST activity to attacker-controlled endpoints and retrieval of multiple DLL files further confirm post-compromise activity and data exfiltration. The combination of encrypted C2 communication, sustained remote access capability, and data exchange patterns indicates that the attacker maintained control over the system and had the capability to collect and exfiltrate sensitive information.

The presence of StealC activity indicates a high risk of credential theft, including browser-stored passwords, session cookies, and system information, which may enable further compromise of user accounts and enterprise services.

### Risk Level: High

This incident represents a high-risk compromise, as it enables unauthorized remote access, persistence, and potential data theft. Immediate containment, host isolation, credential rotation, and full system remediation are required.

This activity should be treated as a confirmed security incident requiring immediate incident response, containment, and remediation actions.